American Computer Science League

2025 Finals ● Short Problems ● Junior Division

1. Boolean Algebra

Which of the following is equivalent to this Boolean expression?

$$\overline{AB} + \overline{A}(B + \overline{A}) + \overline{A + B}$$

A. $A + \overline{B}$
B. $\overline{A} + \overline{B}$
C. $\overline{A} + B$
D. $A + B$
E. None of the above

2. Boolean Algebra

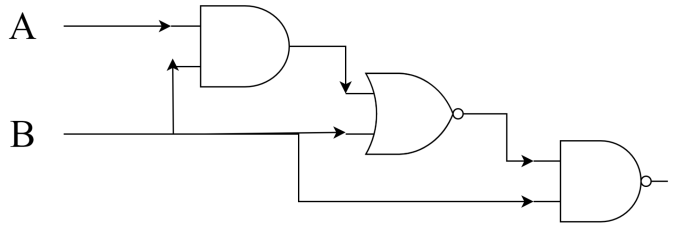
How many ordered triples make the following Boolean expression TRUE?

$$\overline{A}\,\overline{B}\,C + \overline{A}\,B\,\overline{C} + A\,\overline{B}\,\overline{C} + \overline{A\,B\,C}$$

A. 1
B. 4
C. 7
D. 8
E. None of the above

3. Digital Electronics

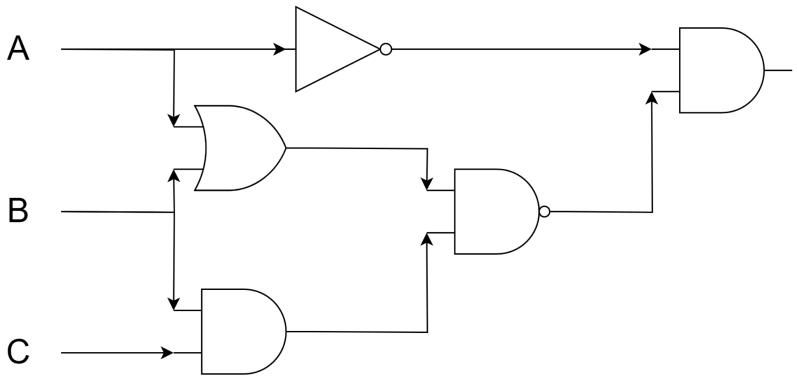
Simplify the Boolean expression for the following digital circuit:



A. 0
B. A
C. B
D. 1
E. None of the above

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 1
B. 3
C. 5
D. 7
E. None of the above

;

5. Bit-String Flicking

Evaluate the following expression:

(LSHIFT-2 01101) OR (RCIRC-3 (10101 AND NOT 01101))

A. 01011
B. 11011
C. 00101
D. 10110
E. None of the above

6. Bit-String Flicking

Solve for X, a bit-string of 5 bits:

(LCIRC-2 X) AND (RSHIFT-1 00110) OR 01010 = (RCIRC-3 11010)

A. *0***
B. 101**
C. *1***
D. 0000*
E. None of the above

7. Recursive Functions

Find $f(17)$, given:

$$f(x) = \begin{cases} f(x-3)+1 & \text{if } x \text{ is odd and prime} \\ f\left(\frac{x}{2}\right)-2 & \text{if } x \text{ is even} \\ 2x & \text{otherwise} \end{cases}$$

Note: remember that 1 is not a prime number

A. -3
B. -2
C. -1
D. 0
E. None of the above

8. Recursive Functions

Find $f(12, 5)$, given:

$$f(x,y) = \begin{cases} f(x-3,\ y+1)-1 & \text{if } x \geq 10 \\ f(x-2,\ y-2)+2 & \text{if } 2 \leq x < 10 \\ xy & \text{if } x < 2 \end{cases}$$

A. 3
B. 4
C. 5
D. 6
E. None of the above

9. Prefix/Infix/Postfix Notation

Translate the following postfix expression to prefix:

a b / a b + c / c 2 ^ − +

A. + − / a b / + a b c ^ c 2
B. + a / b − / + a b c ^ c 2
C. + / a b − + / a b c ^ c 2
D. + / a b − / + a b c ^ c 2
E. None of the above

10. Prefix/Infix/Postfix Notation

Define: a # b = a*b − 2

Evaluate the following prefix expression if all numbers are single digits:

− + # 2 3 * 4 + / 8 2 3 # 4 5

A. 7
B. 14
C. 18
D. 28
E. None of the above

11. Computer Number Systems

Convert 52425_8 to hexadecimal.

A. 5551
B. 5155
C. 5515
D. 1555
E. None of the above

12. Computer Number Systems

Tickets were sold in sequential order for a rodeo starting at $2A_{16}$ and ending at $1A3_{16}$. If the cost per ticket was 1010_2 dollars, what were the gate receipts in decimal?

A. $3760
B. $3770
C. $3780
D. $3790
E. None of the above

13. Data Structures

Build a binary search tree for:

FARMERSMARKET

List the nodes at depth 3 from left to right.

A. A R
B. A E M S
C. E M R T
D. K T
E. None of the above

14. Data Structures

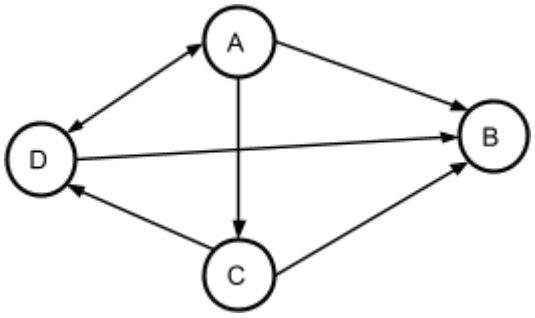
Given an empty queue and the following commands, what will be popped next?

```
PUSH("C")
PUSH("I")
x=POP()
x=POP()
PUSH("L")
PUSH("A")
x=POP()
PUSH("N")
PUSH("T")
PUSH("R")
x=POP()
PUSH("O")
x=POP()
```

A. I
B. L
C. T
D. R
E. None of the above

15. Graph Theory

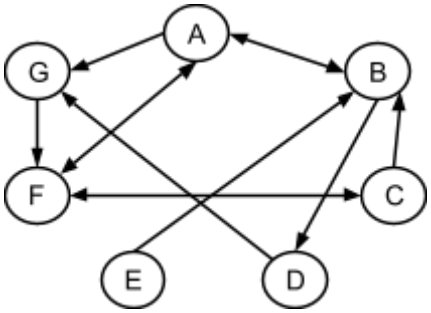
Given the directed graph below, how many paths of length 4 are there?



A. 13
B. 14
C. 15
D. 16
E. None of the above

16. Graph Theory

How many cycles are in the graph below?



A. 4
B. 6
C. 8
D. 10
E. None of the above

17. What Does This Program Do? (Branching)

What is output when the following program is run?

```
x = 21
y = 18
z = 2
if x > y + z then
     x = x + y
end if
if x - z > y then
     y = y - z
end if
output x - y * z
```

A. 7
B. 46
C. 6
D. 42
E. None of the above

18. What Does This Program Do? (Loops)

How many different values are output when the following program is run?

```
for x = 1 to 50 step 3
     if x % 5 == 0 then
          output x
     end if
     if x % 4 == 0 then
          output x
     end if
     s = int(sqrt(x))
     if s*s == x then
          output x
     end if
next x
```

A. 3
B. 5
C. 8
D. 12
E. None of the above

19. What Does This Program Do? (Arrays)

Array **arr** contains 10 integers. How many final values in the array **arr** do not have a value of 1 when this program is run?

2	3	4	5	6	7	8	9	10	11

```
for x = 0 to 8
     for y = x+1 to 9
          if arr(x) != 1 &&
             arr(y) % arr(x) == 0 then
               arr(y) = 1
          end if
     next y
next x
```

A. 4
B. 5
C. 6
D. 7
E. None of the above

20. What Does This Program Do? (Strings)

The function *shift(ch,x)* returns the letter *ch* shifted *x* characters to the left or the right (e.g. shift("A",1) = "B" and shift("Z",-3) = "W"). What is the output for the following program?

```
ans = ""
s = "ACSL_IS_FUN."
for x = 0 to len(s)-1
     if s[x] >= "A" && s[x] <= "Z" then
          if s[x] < "M" then
               ch = shift(s[x],3)
          else
               ch = shift(s[x],-2)
          end if
          ans = ans + ch
     end if
next x
output ans
```

A. LSIQLOQFD
B. DFOQLQISL
C. DFPJGPDTL
D. DFQOLQISL
E. None of the above